6 Data Transfer Schemes

Data Transfer Schemes

1. Programmed data transfer scheme
2. DMA transfer scheme

Programmed Data Transfer Schemes

In programmed data transfer schemes, the data transfer is controlled by the CPU. Data is transferred from an I/O device to the CPU. Programmed data transfer schemes can be of three types.

- Synchronous transfer
- Asynchronous transfer
- Interrupt driven data transfer

Synchronous Transfer

In synchronous data transfer, the device that sends the data and device that receives the data are synchronised with the same clock. This scheme is suitable when both MPU and I/O devices match in speed. The data transfer takes place using IN and OUT instruction in case of I/O mapped I/O devices and with instructions related to memory in case of memory mapped I/O devices.

Asynchronous Transfer

In asynchronous data transfer, the transfer of data is not based on a predetermined timing pattern. This technique is used when the speed of a microprocessor and that of I/O devices is not matching.

Interrupt-driven Data Transfer

In interrupt-driven data transfer, the microprocessor initiates the I/O device to get ready, and then, it executes its main program instead of remaining in loop to check the status of the I/O device. When the I/O device is ready to transfer data, it sends a high signal to the microprocessor through the interrupt line. On receiving an interrupt, the microprocessor completes the current instruction being executed, and then, attends to the I/O device. After completing the data transfer, the microprocessor returns to the main program. Interrupt data transfer is used for slow I/O devices.

Direct Memory Access (DMA) and DMA Controller

The DMA is a process of high speed direct data transfer between memory and external peripherals (I/O device) without programming intervention. DMA involves communication or data transfer controlled by an external peripheral. 8237/8257 ICs are DMA controllers. The DMA controller takes control of the buses and transfers data directly between memory and external peripherals. The 8085 microprocessor has HOLD (Hold) and HLDA (Hold Acknowledge) pins for direct memory access type of I/O communication.

The DMA controller sends a hold request by sending a high signal on the HOLD pin. The processor completes the execution of the current machine cycle; floats (i.e., high impedance state) the address, data and control lines; and also sends the Hold Acknowledge (HLDA) signal.

There are two modes for DMA data transfer.

- Burst mode
- Cycle stealing mode

Burst mode: in this mode of data transfer, the I/O device withdraws the DMA request only after all data bytes have been transferred. This mode transfers a block of data in one go. This technique is used in hard disc drives where data transfer cannot be stopped or slowed down without loss of data.

Cycle stealing mode: in this mode of data transfer, the I/O device withdraws DMA request after transfer of a few bytes and sends the data in intervals. So, the long block of data is transferred in sequence of the DMA cycle. The cycle stealing mode is suitable when a long block of data is to be transferred with minimum interference with microprocessor/MPU activities.

The 8237 DMA Controller

The 8237 is a 40 pin programmable DMA controller. It has four independent channels, which can transfer 64K bytes each independently. It is interfaced with microprocessors and peripherals, such as hard disc or floppy disc. It can operate in two modes called master mode and slave mode. In slave mode, it acts like an I/O device to the microprocessor; and in master mode, it acts like a data transfer processor to peripherals.

Programmable Peripheral Interface (8255 A)

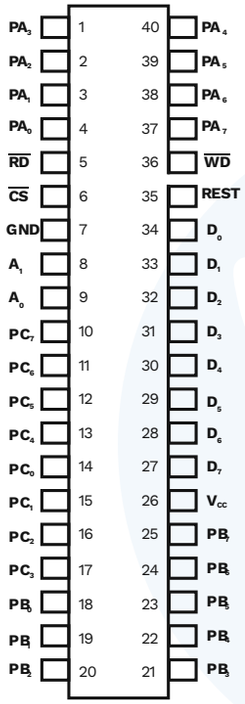


Fig. 6.1 Pin Configuration of 8255A

The 8255A is a 40 pin programmable peripheral interface used for data transfer with simple I/O or interrupt-driven I/O ports—A B and C. The eight bits of port C can be used as individual bits or be divided into two 4-bit ports C_{upper} (C_U) and C_{Lower}(C_l)

Control Signals

The 8255 has six control signals used to perform various functions.

RD (Red): This control signal enables the read operation. When this signal is low, the MPU reads data from a selected I/O port of the 8255A.

WR (Write): This control signal enables the write operation. When this signal goes low, the MPU writes data into the selected I/O port or the control register.

RESET (Reset): This is an active high signal. It clears the control register and sets all ports in the input mode.

CS, A_0 and A_1: These are device select signals. CS is connected to a decoded address and A_0, and A_1 generally connected to MPU address lines A_0 and A_1, respectively.

The CS signal is the master chip select, and A_0 and A_1 specify one of the I/O ports or the control register as given below.

CS	A_0	A_1	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control register
1	X	X	8255 is not selected

Control Word and Control Register

The 8255 can be operated in two modes called Bit Set/Reset (BSR) mode and I/O mode. The BSR mode is used to set or reset the bits in port C. The I/O mode is further divided into three modes called Mode 0, Mode 1, and Mode 2.

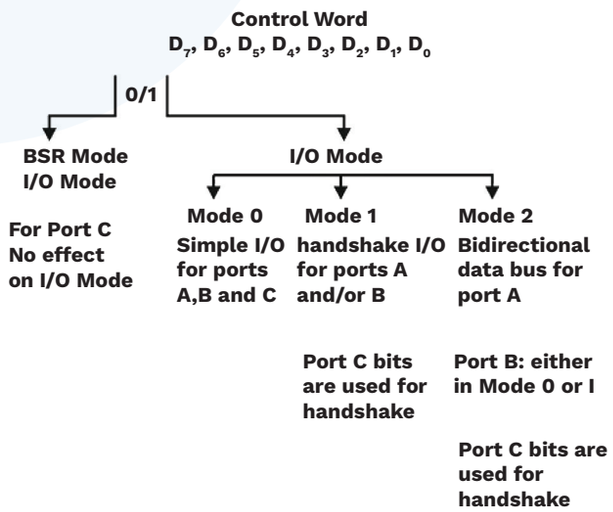


Fig. 6.2 Modes of 8255A

I/O Modes and Its Control Word

The I/O mode is selected when bit D_7 of the control register is 1. The remaining bits D_6-D_0 determine the I/O function in various modes as shown in the figure.

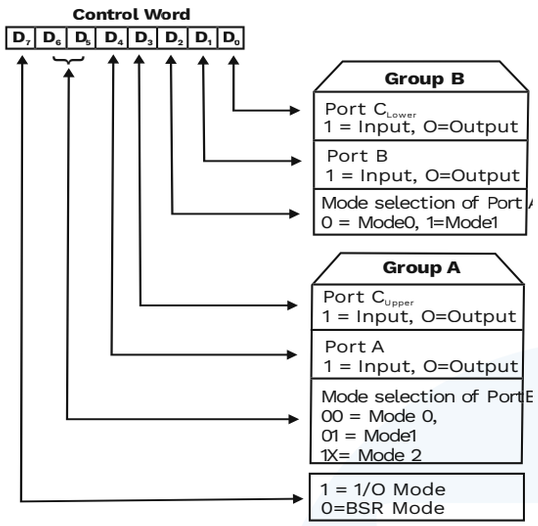


Fig. 6.3 Structure of Control Word

BSR Mode and Its Control Word

The BSR mode is selected when bit D_7 of the control register is 0. This mode sets or resets bits of port C. The BSR control word does not affect the functions of ports A and B.

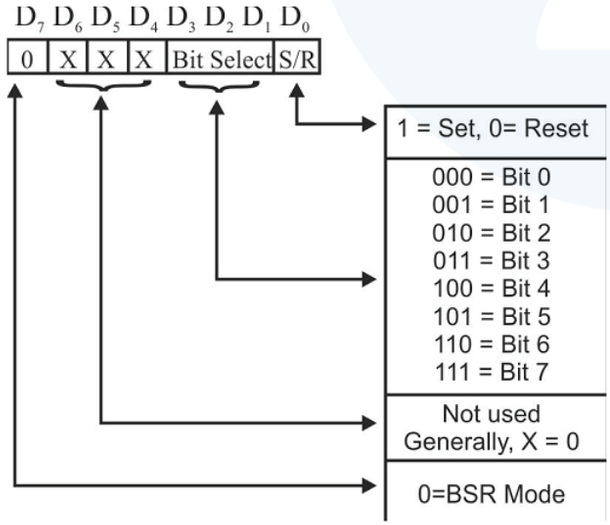


Fig. 6.4 Control Word for BSR Mode

8155 Multipurpose Programmable Device

The 8155 is a multipurpose programmable device specifically designed to be compatible with the 8085 microprocessor. It has two functionally independent sections: the first is 256 bytes of R/W memory, and the second is a programmable I/O.

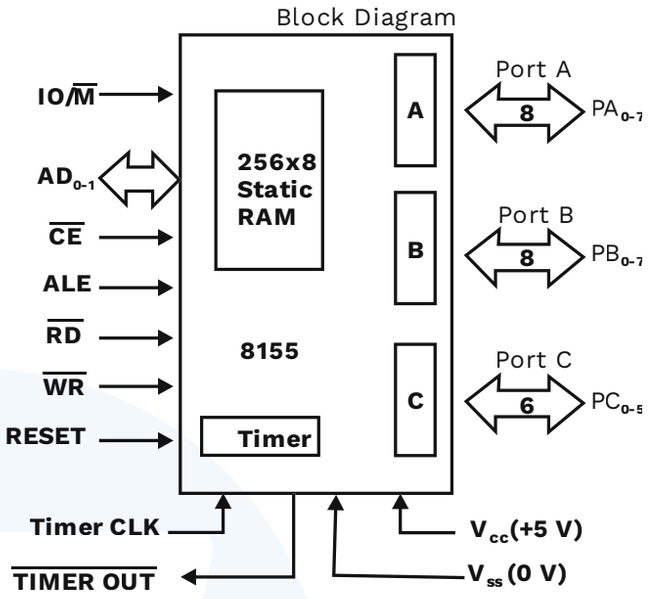


Fig. 6.5 Block Diagram of 8155

Control Signals

The 8155 has six control signals called CE, IO/M, ALE RD, WR and RESET. All control signals except the Chip Enable (CE) are directly generated by 8085.

1. CE (Chip Enable): this signal enables the chip to perform the operation. It is connected to the decoded high order bus of the microprocessor.
2. IO/M: when this signal is low, the memory selection is selected, and when it is high, the I/O section (including timer) is selected.
3. ALE (Address Latch Enable): this signal latches the low-order address AD_7 – AD_0, CE and IO/M into the chip.

8254 (8253) Programmable Interval Timer

The 8254 is a 24-pin programmable interval timer/counter. It generates accurate time delays and can be used for applications, such as a real-time clock, an event counter, a digital one-shot, a square-wave generator, and a complex waveform generator.

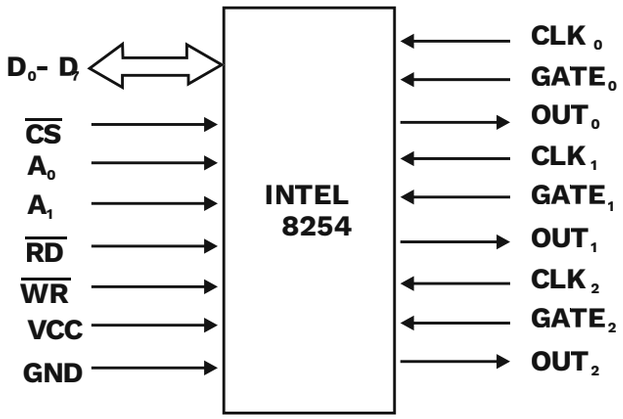


Fig. 6.6 Schematic Diagram of Intel 8254

RD (Read): it is an active low signal. When this pin is low, the CPU reads data from the counter.

WR (Write): it is an active low signal. When this pin is low, the CPU outputs data in the form of mode information or loading of counters.

A_0, A_1: these pins are connected to the address lines A_0 and A_1 of the address bus. These are used to select one of the three counters. These are also used to address the control word registers for mode selection.

CS: Chip Select

D_0-D_7: Bidirectional data bus

CLK_0, CLK_1, and CLK_2 are clocks for Counter 0, Counter 1 and Counter 2, respectively.

$GATE_0$, $GATE_1$, and $GATE_2$ are gate terminals of Counter 0, Counter 1, and Counter 2, respectively.

OUT_0, OUT_1 and OUT_2 are output terminals of Counter 0, Counter 1 and Counter 2, respectively.

Block Diagram

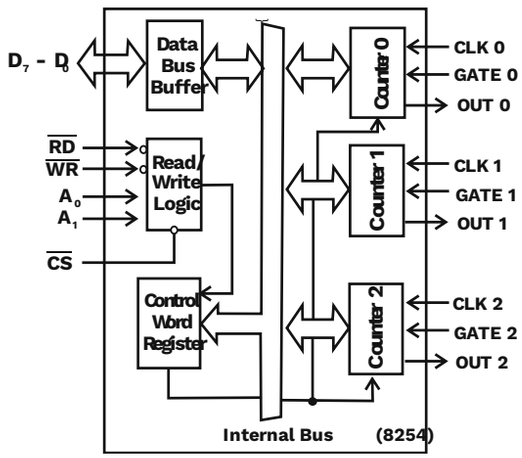


Fig. 6.7 Block Diagram of 8254

Control Section

The control section of 8254 timer has five signals called RD (Read), WR (Write), CS (Chip Select), and the address line A_0 and A_1.

CS	A_1	A_0	Section
0	0	0	Counter 0
0	0	1	Counter 1
0	1	0	Counter 2
0	1	1	Control Register
1	X	X	8254 not enable

Control Word and Control Register

The control word decides the selection of the counter, its mode of operation, loading, sequence of the count, and selection of binary or BCD counting.

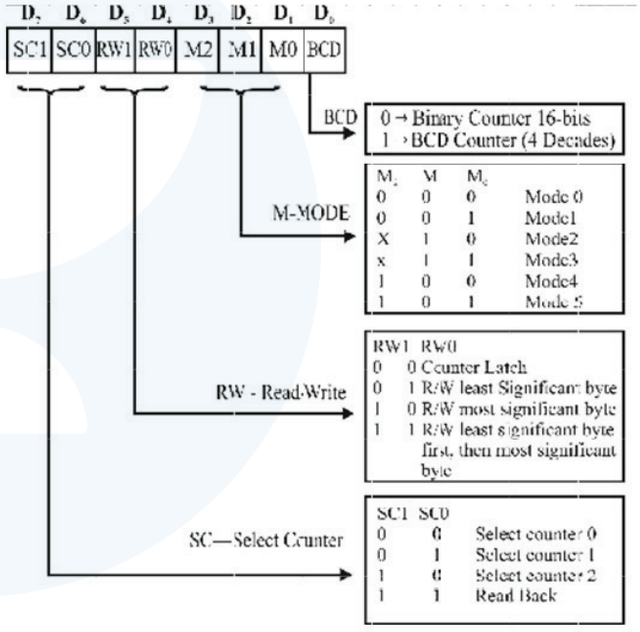


Fig. 6.8 Control Word Format of 8254

The 8259A Programmable Interrupt Controller

The programmable interrupt controller is used for interrupt-driven data transfer between the I/O device and microprocessor. It is used when many I/O devices are to be connected to the same interrupt line. The Intel 8259A is a 28 pin single chip programmable interrupt controller.

PIN Configuration of 8259

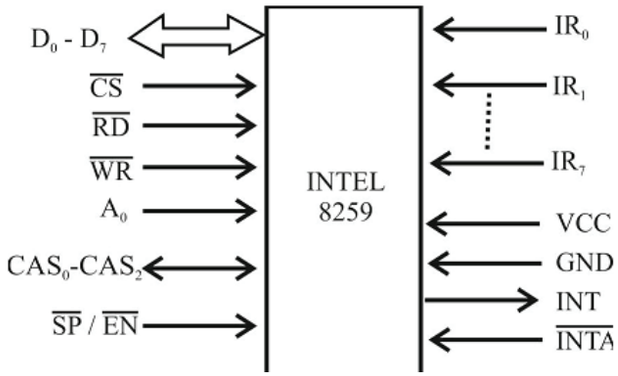


Fig.6.9 8259 Interrupt Controller

CS It stands for Chip Select. This signal enables the chip to perform various functions.

WR It is an active low signal. When the signal on this pin is low, 8259 accepts command word form microprocessor.

RD It is an active low signal. When the signal on this pin is low, 8259 sends various status signals on the data bus for the microprocessor.

D_0-D_7 It is a bidirectional data bus control status and interrupt vector information are transferred via this bus.

CAS_0 - CAS_2 These are cascade lines.

INT This line is used to interrupt the microprocessor

INTA It stands for Interrupt Acknowledge. It is connected to the INTA of the microprocessor.

IR_0-IR_7 These lines are used for interrupt requests. I/O devices send interrupt requests through these lines.

A_0 This address line acts in conjunction with RD, WR, and CS to interpret command words the CPU writes and status the CPU wants to read.

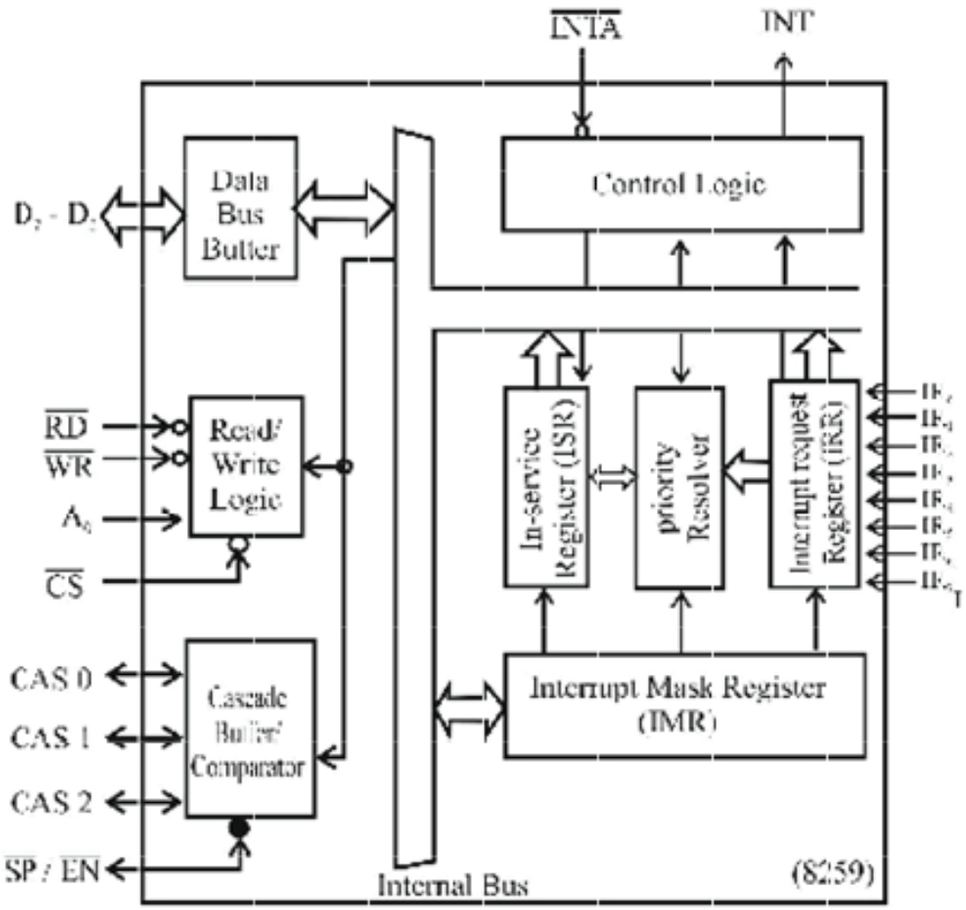


Fig. 6.10 Block Diagram of 8259

Practice Exercise

Level I

1. Microprocessor 8085 regains control of the bus:
(A) immediately after HOLD goes low
(B) immediately after HOLD goes high
(C) after half-clock cycle after HLDA goes high
(D) after half-clock cycle after HLDA goes low

2. Consider a system consisting of a microprocessor, a memory, and peripheral devices connected by a common bus. During DMA data transfer, the microprocessor:
(A) only reads form the bus
(B) only writes to the bus
(C) both reads from and writes to the bus
(D) neither reads form nor writes to bus

3. If 8255 A chip is selected when A_2 to A_7 pins are 1, what is the address of Port C?
(A) FC
(B) FD
(C) FB
(D) FE

4. If 8255 A chip is selected when A to A bits are all 1, what is the address of Port A?
(A) 80
(B) FA
(C) FB
(D) FC

5. What must be the contents of the control world of Intel 8255 for mode 0 (operation) and for the following ports configuration.
Port A- output, Port B- output,
Port C_{lower} – output, Port C_{upper}- input
(A) 85H
(B) 86 H
(C) 87 H(D)
(D) 88 H

6. Consider the following statements.
The advantages of cycle stealing in DMA are as follows.
I. It increases the maximum I/O transfer rate.
II. It reduces interference by the DMA controller in CPU'S memory access.
III. It is beneficially employed for I/O device with shorter bursts of data transfer.
Which of the statements given above are correct?
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III only

7. Which of the following is not a characteristic of a transparent DMA mode of I/O operation?
(A) The external logic steals cycles from the CPU.
(B) The normal rate of execution is slowed down.
(C) Only one word can be transferred at a time.
(D) Data are transferred to/from memory directly.

8. **Assertion (A):** DMA is faster than either interrupt initiated I/O or polling based I/O for very large data transfer.
Reason (R): DMA takes control of the system buses and needs no processor intervention during data transfer.
(A) Both A and R are individually true and R is the correct explanation of A.
(B) Both A and R are individually true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

9. The Direct Memory Access channel facilitates data to move into and out of the system:
(A) without subroutine.
(B) with equal time delay.
(C) without program intervention.
(D) on first come first serve basis.

10. An 8254 programmable interval timer consists of independent 16-bit programmable counters. This number is:

(A) 2
(B) 3
(C) 4
(D) 5

11. Match List I (chips related to microprocessors and controllers) with List II (their function) and select the correct answer using the codes given below the lists.

List I

a. 8255A
b. 8254
c. 8259
d. 8096

List II

I. Interrupt controller
II. Microcontroller
III. Counter
IV. Peripheral interface

Codes:

	a	b	c	d
(A)	III	IV	II	I
(B)	IV	III	II	I
(C)	IV	III	I	II
(D)	III	IV	I	II

Level II

1. The interfacing device used for the generation of accurate time delay in a microcomputer system is:
(A) Intel–8251
(B) Intel–8257
(C) Intel–8253
(D) Intel–8259

2. What is the purpose of a DMA facility in a microprocessor based system?
(A) To increase the speed of data transfer between the μP and the I/O devices
(B) To increase the speed of data transfer between the μP and the memory
(C) To increase the speed of data transfer between the memory and the I/O devices
(D) To improve the reliability of the system

3. Handshaking mode of data transfer is:
(A) synchronous data transfer
(B) asynchronous data transfer
(C) interrupt driven data transfer
(D) level mode of DMA data transfer

4. In mode '0' (zero) operation of 8255, which of these can be used as port?
(A) A as input port only
(B) B as output port only
(C) A as output only
(D) A as input or output only

5. **Assertion (A):** Port B of 8255 A can only be used as input port.
Reason (R): Port B can only be operated in Mode 0 and Mode 1
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

6. The 8259 A programmable interrupt controller can:
I. manage eight interrupts.
II. vector an interrupt request anywhere in the memory map.
III. have an 8-bit or a 16-bit interval between interrupt vector locations.
IV. be initialised with operational command words (OCW's).
(A) I and II
(B) II and III
(C) III and IV
(D) I, II, III, and IV

7. Which of the following methods could be used for I/O operation in an 8085 microprocessor?
I. Handshaking
II. Interrupting
III. Direct Memory Access
Select the correct answer using the codes given below.
(A) I and II
(B) I and III
(C) II and III
(D) I, II, and III

8. When a processor uses handshaking, the following handshaking signal are present.

I. Acknowledge
II. Data transfer
III. Status
IV. Strobe

What is the correct sequence in which the signal appears?

(A) Strobe–Status–Data Transfer–Acknowledge
(B) Status–Strobe–Data Transfer–Acknowledge
(C) Status–Data Transfer Acknowledge–Strobe
(D) Strobe–Data Transfer–Acknowledge–Status

9. Which of the following statements is correct? The I/O Method used in a microcontroller is:

(A) handshaking
(B) interrupting
(C) Direct Memory Access
(D) All of the above

10. With regard to DMA operation, which of the following statements is incorrect?

(A) DMA operation begins with DMA controller sending a DMA request to processor.
(B) DMA operation ends with DMA controller sending an interrupt to processor.
(C) DMA controller has a bidirectional address bus.
(D) DMA controller has bidirectional READ and WRITE control lines.

Answer Key

Level I

1. (A) **2.** (D), (A) **3.** (D) **4.** (D) **5.** (D) **6.** (A) **7.** (B) **8.** (A) **9.** (C) **10.** (B)
11. (C)

Level II

1. (C) **2.** (C) **3.** (B) **4.** (D) **5.** (D) **6.** (A) **7.** (D) **8.** (D) **9.** (D) **10.** (B)